

AI Use Cases

Report Designed and Developed by: AIM ASEAN Programme

Remarks: This use cases report is designed to provide valuable insights and data for the implementation of AI strategies and digital transformation in the ASEAN region. Implementers are encouraged to extract relevant findings from this report to inform their initiatives, ensuring that all sources are properly cited. When utilizing the content, it is important to adapt the research findings to the local context and update any data as necessary to reflect the most current trends and developments. All sources referenced in this report should be credited accordingly to maintain academic integrity and uphold the credibility of the research.

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Energy

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/asean-energy-cooperation>

Use Case: Digital Twin for Grid Management at Tenaga Nasional Berhad (TNB), Malaysia



How AI Helps

TNB uses AI to create a "**digital twin**," a virtual replica of its grid infrastructure. This AI-powered model analyzes real-time data from sensors, smart meters, and historical records to **predict when and where equipment, like cables, might fail**. It also helps optimize the flow of electricity from various sources, including intermittent renewables like solar.

The Problem

TNB, Malaysia's largest electricity utility, faces challenges with aging infrastructure, unplanned outages, and the integration of renewable energy sources. This results in significant operational costs and service disruptions.

Source: [TNB](#)

Business Impact

The predictive maintenance enabled by the digital twin has significantly reduced unplanned outages, improved grid reliability, and optimized asset management. This leads to lower operational costs and enhanced customer satisfaction.

Use Case: AI for Renewable Energy Forecasting at National Grid Corporation (NGCP), Philippines



How AI Helps

NGCP uses an **AI-powered forecasting system** that analyzes a vast amount of data, including real-time weather patterns, satellite imagery, historical energy consumption, and solar irradiance levels. The AI models can **predict solar and wind energy output** with a high degree of accuracy, giving NGCP advance notice of potential energy shortfalls or surpluses. This allows the grid operator to plan ahead, adjust other power plant outputs, or activate energy storage systems to maintain a stable and reliable power supply.



The Problem

The Philippines is an archipelago with a growing number of intermittent renewable energy sources, such as solar and wind farms, integrated into its grid. The unpredictable nature of these sources—due to fluctuating weather—makes it difficult for NGCP to maintain grid stability and balance supply and demand. This can lead to power outages or the need to use more expensive, fossil-fuel-based "peaker plants" to compensate.

Source: [Reglobal](#)



Business Impact

The predictive maintenance enabled by the digital twin has significantly reduced unplanned outages, improved grid reliability, and optimized asset management. This leads to lower operational costs and enhanced customer satisfaction.

Use Case: AI-Driven Building Management System (BMS) at Marina One, Singapore



✧ How AI Helps

Marina One has deployed an advanced, **AI-driven BMS** that integrates with a network of IoT sensors throughout the building. The AI analyzes real-time data on occupancy, temperature, humidity, and CO₂ levels within different zones. Based on this data, the system's machine learning algorithms **autonomously adjust climate control and lighting** to optimize for both occupant comfort and energy efficiency. For example, it can dynamically reduce cooling in unoccupied areas or dim lights in conference rooms when CO₂ levels indicate no one is present.

📖 The Problem

Large-scale commercial developments, like Marina One, face the dual challenge of high energy consumption and the need to meet strict sustainability standards, such as Singapore's Green Mark. Inefficient and static building management systems (BMS) lead to significant energy waste, particularly from HVAC and lighting, which are major operational costs.

Source: [Venturous Group](#), [Cyrill \(Pexels\)](#)

⚙️ Business Impact

The AI-driven system has enabled Marina One to achieve substantial, quantifiable energy savings. The building's energy performance is up to **35% more efficient** than that of a typical commercial building. This not only significantly reduces operational costs but also helps the development achieve its sustainability goals and maintain its prestigious Green Mark Platinum certification, enhancing its market value and tenant appeal.

Use Case: Oil and Gas Pipeline Integrity

The Problem

Maintaining the integrity of thousands of miles of oil and gas pipelines is a major challenge. Issues like corrosion, leaks, and structural faults can lead to costly repairs, environmental disasters, and safety hazards.

Traditional inspection methods are often manual, time-consuming, and cannot provide continuous monitoring, which means problems are often discovered only after they have escalated.

How AI Helps

AI utilizes data from a network of sensors (IoT) along pipelines that monitor variables like pressure, temperature, and flow rate. It also analyzes data from visual inspections, including satellite imagery and drone footage. AI algorithms, particularly those for **anomaly detection**, can process this data in real-time to predict potential failures before they occur. For example, a minor drop in pressure or a subtle change in metal integrity detected by an AI model can trigger an alert for a proactive inspection and repair.

Business Impact

This leads to a shift from reactive to proactive, predictive maintenance. The most significant benefits are enhanced safety for workers and surrounding communities and a dramatic reduction in the risk of environmental spills. It also translates to significant cost savings by preventing catastrophic failures and minimizing downtime, as maintenance can be scheduled efficiently before an outage occurs.

Use Case: Power Grid Management

The Problem

The power grid faces significant challenges with the integration of intermittent renewable energy sources like wind and solar.

Traditional, static grid management systems struggle to handle the variable supply and demand, leading to inefficiencies, potential blackouts, and grid instability. A key issue is accurately forecasting energy production from renewables and balancing it with real-time consumer demand.

How AI Helps

AI-powered systems, often referred to as "smart grids," use machine learning algorithms to analyze vast amounts of data from IoT sensors, smart meters, and weather forecasts. These models can accurately **predict energy supply** from renewable sources and forecast demand across different regions and times. The AI then dynamically **adjusts power distribution and manages energy storage systems**, such as large-scale batteries, to maintain a stable balance between supply and demand in real-time.

Business Impact

This translates to greater grid stability and reliability, reducing the risk of blackouts. It also enables the efficient integration of more renewable energy, helping utilities meet clean energy goals while minimizing the need for costly, on-standby fossil fuel plants. By optimizing energy distribution and reducing waste, AI leads to lower operational costs and improved energy efficiency.

Use Case: Renewable Energy Optimization

The Problem

A key challenge for renewable energy, especially solar and wind, is their inherent variability. Their output depends on unpredictable weather conditions, which makes it difficult to reliably plan energy supply.

This "intermittency" can lead to energy waste during periods of high production and energy shortages when production is low.

How AI Helps

AI systems use advanced analytics and machine learning to create highly accurate predictive models. These models combine historical weather data with real-time forecasts (e.g., cloud cover, wind speed) to **predict with precision how much energy a solar farm or wind turbine will produce**. The AI then integrates this forecast with real-time energy demand data. It can then make real-time decisions, for example, to adjust the angle of a wind turbine's blades for maximum energy capture or to divert excess solar energy to a storage facility.

Business Impact

This application of AI directly addresses the intermittency issue, leading to increased energy production and efficiency. It allows energy producers to maximize the value of their renewable assets by accurately forecasting and committing to energy delivery schedules, which can lead to higher profits in energy markets. It also helps to reduce energy waste and makes renewable energy a more reliable and profitable source, accelerating its adoption as a primary power source.

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Minerals

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/asean-minerals-cooperation/>

Use Case: AI-powered Mine Operations at BHP, and its BMA Joint Venture in Queensland



How AI Helps

BHP uses AI and digital twins to create **virtual replicas of its mining value chains**. The AI analyzes vast amounts of data from equipment, sensors, and historical plans to predict production outcomes, identify bottlenecks, and optimize processes in real time. For example, AI can determine the best route for trucks or the optimal number of loading units to operate.

The Problem

Mining operations are highly complex, with numerous moving parts from excavation to transportation. Optimizing every decision manually is impossible, leading to inefficiencies, increased costs, and safety risks.

Source: [BHP](#)

Business Impact

AI has led to a 10% increase in productive movement at one of BHP's autonomous operations. By testing "what-if" scenarios in the digital twin, the company saves significant time and cost and improves safety by reducing the need for on-site trials.

Note: BHP operates in Southeast Asia and the broader APAC region

Use Case: AI for Fatigue Detection at BHP

BHP

The Problem

BHP, a global mining giant, faced a persistent and dangerous problem with driver fatigue in its operations. Operating massive 400-ton haul trucks, fatigued drivers posed a significant safety risk, leading to accidents and operational downtime. The traditional method of relying on drivers' self-reporting or simple eye-tracking sensors was unreliable and often too late.

Source: [BHP](#)

How AI Helps

BHP implemented a new AI-powered system that uses specialized caps with sensors to **monitor the brainwaves of truck drivers**. These sensors, fitted within the driver's helmet, continuously analyze brainwave patterns to detect early signs of fatigue. When the AI system identifies a potential risk of a driver falling asleep, it immediately alerts the driver and, crucially, also sends a notification to a back-office supervisor. This allows for proactive intervention to remove the driver from the hazardous situation before an accident can occur.

Business Impact

The implementation of this technology has had a profound impact on safety and efficiency. A key outcome reported by BHP is zero accidents due to driver drowsiness since the system's deployment. Beyond the immeasurable value of preventing injuries and saving lives, this has also led to significant cost savings by avoiding accidents, minimizing downtime, and improving overall productivity. The company described it as a "much better technology" that could save millions in the future.

Note: BHP operates in Southeast Asia and the broader APAC region

Use Case: US Steel Boosts Maintenance Efficiency with Generative AI



The Problem

At its Minnesota iron ore operations, US Steel faced a common but costly challenge: equipment maintenance. Technicians spent valuable time manually searching through thousands of pages of technical manuals, standard operating procedures (SOPs), and complex diagrams to diagnose issues and find solutions. This process was inefficient, led to delays in repairs, and increased the risk of errors, ultimately raising maintenance costs and impacting productivity.

Source: [mine](#)

How AI Helps

To solve this, US Steel partnered with Google Cloud to deploy **MineMind**, a **generative AI-powered application**. MineMind acts as a smart consultant, allowing maintenance teams to instantly query the system for answers to their technical questions. For example, a technician can simply ask, "How do I troubleshoot a hydraulic system malfunction?" or "What's the procedure for calibrating a specific sensor?" In response, the AI retrieves relevant SOPs and provides detailed summaries and diagrams. It also includes a validity score and source references, ensuring the information is accurate and trustworthy. This eliminates the need for technicians to manually sift through extensive documents.

Business Impact

The implementation of MineMind has delivered significant results. Initially focusing on A60 haul trucks, US Steel estimates the tool reduces the time required to complete work orders by approximately 20%. This enhancement in productivity directly translates into reduced maintenance costs and improved operational efficiency. The success of this pilot program has led US Steel to plan the development of more generative AI applications to address high-impact areas across its entire value chain.

Use Case: Mineral Exploration and Discovery

The Problem

Mineral exploration is a high-risk and expensive process with a low success rate. Traditionally, geologists rely on a limited number of data points from geophysical surveys and drilling to make decisions about where to explore.

This approach is time-consuming, costly, and can easily miss valuable deposits hidden in complex geological formations.

How AI Helps

AI and machine learning algorithms can **analyze and integrate vast, multi-modal datasets** that are too complex for humans to process. These datasets include historical exploration data, satellite imagery, geophysical and geochemical surveys, and drilling logs. AI models can identify subtle patterns, anomalies, and correlations across this data that indicate the presence of mineral deposits. For example, AI can predict the location of ore bodies with greater accuracy by finding relationships between different data types, significantly reducing the amount of unproductive drilling.

Business Impact

This translates to reduced exploration costs and timelines by minimizing unnecessary drilling and fieldwork. It also leads to a higher discovery success rate, enabling companies to find new, economically viable deposits more efficiently. By making data-driven decisions, companies can allocate resources more effectively and reduce the overall financial risk associated with exploration.

Use Case: Predictive Maintenance and Fleet Management

The Problem

Mining operations rely on a fleet of heavy-duty and expensive machinery, such as haul trucks, excavators, and crushers. Unexpected equipment failures can lead to significant and costly downtime, production losses, and safety hazards.

Traditional, time-based maintenance schedules are often inefficient, as they can either lead to premature servicing or fail to prevent a breakdown.

How AI Helps

AI-powered predictive maintenance systems use sensors (IoT) on equipment to collect real-time data on their performance, including vibration, temperature, and fluid levels. Machine learning algorithms analyze this data to **predict when a component is likely to fail**. The system can then alert maintenance teams to perform repairs during planned downtime, before a breakdown occurs. Additionally, AI is used to **optimize the routes of autonomous haulage trucks**, minimizing fuel consumption and wear and tear.

Business Impact

The primary benefit is a shift from reactive to proactive maintenance, which significantly reduces unplanned downtime and associated costs. This increases productivity and extends the lifespan of expensive machinery. Furthermore, the use of autonomous, AI-optimized fleets improves operational efficiency, reduces fuel consumption, and enhances worker safety by removing personnel from hazardous environments.

Use Case: Mine Safety and Hazard Monitoring

The Problem

The mining environment is inherently dangerous, with risks ranging from rockfalls and gas leaks to equipment collisions.

Monitoring these hazards manually is difficult and can expose workers to danger. Real-time, comprehensive monitoring is crucial for preventing accidents and ensuring worker safety, but it is a monumental task.

How AI Helps

AI systems utilize a network of sensors, cameras, and drones to continuously monitor the mine environment. Computer vision algorithms can **analyze video feeds to detect unsafe conditions**, such as workers entering a restricted zone, improper use of personal protective equipment, or the early signs of a structural collapse. AI-powered sensors can also **monitor for changes in air quality, gas levels, and ground stability**, automatically triggering alarms and alerts to notify workers of an imminent hazard. Wearable IoT devices equipped with AI can **track a worker's health and location**, and even detect fatigue or a fall, providing real-time alerts to supervisors.

Business Impact

This leads to a dramatic improvement in mine safety and a reduction in workplace accidents. By providing real-time hazard detection and automated alerts, AI enables a rapid response to dangerous situations, potentially saving lives. The data collected by these systems also helps companies identify high-risk areas and implement more effective safety protocols, leading to a safer, more compliant, and more productive workforce.

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Food, Agriculture and Forestry

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/asean-minerals-cooperation/>

Use Case: Monsoon Prediction for Indian Farmers

The Problem

More than 100 million Indian farmers depend on the summer monsoon to decide when to sow seeds, apply fertiliser, and hire seasonal labour.

The timing and intensity of the rains have become increasingly erratic because of climate change, and the government's conventional forecasting models often give only broad regional guidance or arrive too late. Inaccurate or delayed predictions can lead to crop failure, wasted inputs, and financial losses, especially for smallholders who operate on thin margins.

Source: [Google Research Blog](#)

How AI Helps

Google Research built a deep-learning climate model known as **NeuralGCM** that fuses decades of historical weather data with current satellite observations and global climate reanalysis. The model captures complex atmospheric interactions and **provides highly localised forecasts of monsoon onset up to two weeks earlier than traditional systems**. Google partners with India's Ministry of Agriculture and local agri-tech platforms to deliver the predictions directly to farmers' mobile phones in local languages via SMS and smartphone apps, ensuring actionable insights even in rural areas with limited internet connectivity.

Business Impact

Earlier and more precise rainfall forecasts have reached an estimated 38 million farmers, allowing them to adjust sowing schedules, optimise fertiliser use, and plan irrigation more efficiently. By reducing the risk of crop failure and input wastage, farmers have reported higher net incomes and improved food security. At a macro level, better monsoon forecasting also stabilises supply chains for staples such as rice and pulses, helping local markets and national food reserves remain more resilient to climate variability.

Use Case: SeeTree: Per-Tree Monitoring for Growers and Forest Managers

The Problem

Commercial orchards and forest-management groups often oversee thousands of hectares where the health of each individual tree can determine overall yield and long-term sustainability.

Traditional scouting requires workers to manually inspect trees for pests, disease, or drought stress, a process that is slow, labour-intensive, and prone to human error.

Delayed detection frequently means lower yields, greater chemical use, and missed opportunities for early intervention or for verifying carbon credits tied to healthy forest growth.

Source: [Google Cloud Customer Story](#)

How AI Helps

Operating on Google Cloud's Vertex AI and BigQuery, SeeTree **integrates high-resolution drone imagery, satellite data, and ground-based IoT sensors.**

Computer-vision and machine-learning models automatically identify signs of nutrient deficiency, disease, or water stress at the level of a single tree.

Predictive analytics then estimate per-tree yield and recommend targeted actions, while scalable cloud processing allows near-real-time updates across entire plantations. The platform's architecture lets forest managers and fruit growers monitor millions of trees with minimal human oversight.

Business Impact

By replacing slow manual scouting with AI-driven insights, SeeTree customers have shortened carbon-credit verification and yield-analysis cycles from many months to just weeks. This accelerates revenue from verified carbon offsets and enables growers to act on early disease warnings, cutting pesticide use and improving overall harvest quality. The result is lower operational cost, more sustainable resource management, and the ability to expand production without proportionally increasing labour, a critical advantage in markets facing chronic farm-worker shortages.

Use Case: Forest-Risk Commodity Sourcing with Google Earth Engine

The Problem

Global food and consumer-goods companies that buy commodities like palm oil, cocoa, and soy face growing regulatory and consumer demands to prove that their supply chains are free from deforestation. Monitoring thousands of suppliers across multiple continents is logistically daunting.

Traditional audits rely on infrequent field visits and self-reported data, which are costly, slow, and often unreliable, exposing companies to reputational damage and legal penalties.

Source: [Google Cloud Geospatial Sustainability Case](#)

How AI Helps

Using **Google Earth Engine**, geospatial scientists and corporate sustainability teams apply machine-learning classification to streams of satellite imagery to **detect changes in forest cover in near real time**. The system integrates deforestation alerts with supply-chain maps so that procurement teams can instantly see whether a supplier's farm overlaps with newly cleared land. Companies can then suspend purchases, demand remediation, or work with local partners to halt further clearing. Because the models continuously retrain on new imagery, detection accuracy improves over time.

Business Impact

This AI-driven monitoring enables companies to demonstrate **compliance with regulations such as the EU Deforestation Regulation** and to meet voluntary ESG commitments without the prohibitive costs of constant on-site inspections. Firms can act on alerts within days rather than months, protecting brand reputation and reducing the risk of import bans or fines. Over time, the system lowers the cost of sustainable sourcing, strengthens relationships with environmentally responsible suppliers, and contributes directly to global forest-conservation goals.

Use Case: AI Forecasting to Cut Food Waste at Coop



How AI Helps

Using Google Cloud's AI-powered Data Cloud, Coop built advanced demand-forecasting models that factor in seasonality, promotions, and supply-chain constraints. Scalable compute lets them train multiple models, run real-time simulations, and push accurate forecasts directly into ordering and inventory systems.



The Problem

Coop, a major European grocery chain, was losing money and goodwill to overstocking and spoilage. Legacy on-premise forecasting systems couldn't keep up with fast-changing shopper behaviour, seasonal swings, and the scale of a modern retail network, driving unnecessary food waste and higher costs.



Business Impact

Coop now keeps shelves stocked with fresher products while slashing waste and associated disposal costs. More accurate forecasts mean leaner inventories, lower operating expenses, and a stronger sustainability story, delivering higher customer satisfaction and measurable savings across the entire retail network.

Source: [Sustainable Future](#)

Use Case: Community Forest Monitoring in the Peruvian Amazon

The Problem

Indigenous and rural communities in Peru's Amazon faced accelerating illegal logging and forest loss. Manual patrols over vast areas were slow and expensive, leading to delayed detection and limited deterrence.

How AI Helps

Satellite imagery was analysed with automated change-detection algorithms to create **near-real-time forest-loss alerts**. These alerts were delivered to trained community monitors via mobile devices, allowing rapid verification and targeted patrols.

Business Impact

The AI alerts cut deforestation detection time from weeks to just days, enabling communities to document evidence quickly, improve enforcement, maintain sustainable-harvest certifications that support eco-timber sales, and ultimately strengthen local livelihoods while reducing forest loss compared with control areas.

Source: [Cappello et al., Remote Sensing 2022](#)

Use Case: Smart Rice-Farm Water Management in Thailand

The Problem

Rice farmers in Suphan Buri province faced unpredictable rainfall and high irrigation costs. Over- or under-watering reduced yields and increased pumping expenses.

How AI Helps

An on-farm network of soil-moisture and climate sensors fed data into a cloud platform using machine-learning models to predict irrigation needs. Farmers received automatic alerts and could remotely control pumps through a mobile interface.

Business Impact

The project achieved about a 30% reduction in water use, lowering energy costs for pumping, while farmers reported 5–10% yield increases from more consistent soil conditions, creating a scalable model for other smallholder groups seeking low-cost precision agriculture.

Source: [Suanpang et al.](#)

Use Case: Smart Plant Doctor

The Problem

Walking through your fields, checking every plant takes too much time. By the time you spot a sick plant, the disease has already spread to many others. This can cost you thousands of dollars in lost crops.

How AI Helps

Think of AI like having a **smart camera** that never gets tired. You can put cameras on a drone or set them up in your fields. These cameras take pictures of your plants every day and **look for early signs of sickness** that your eyes might miss.

The AI learns what healthy plants look like on your specific farm. When it spots something wrong, it **sends you a message on your phone** explaining exactly what disease it found and what you should do about it. It even **tells you which spray to use** and when to apply it.

Business Impact

Farmers who use this system usually can save their crops that would have been lost to disease, and how much is saved will depend on how early the detection is going to be, followed by how infectious the plant disease can be. For example, if a wheat farmer caught a fungus problem a few weeks early, he can save most of his crop. **The system potentially will have paid for itself after preventing just one major disease outbreak.**

Use Case: Smart Health Monitoring for Chicken and Pig Farms

The Problem

In chicken and pig farms, disease spreads very fast when hundreds or thousands of animals live close together. You cannot watch every animal closely, and by the time you notice sick animals, the disease has often spread to many others. This makes treatment expensive and can cause you to lose many animals.

How AI Helps

AI cameras and **microphones** watch and listen to your animals **24 hours a day**. For chickens, the system spots sick birds by watching how they move, eat, and behave. For pigs, it listens for coughing sounds that show respiratory disease before you can hear the problem yourself.

When the AI detects sick animals, it **sends alerts to your phone** immediately. The system learns what normal behavior looks like for your specific farm and gets better at spotting problems over time.

Business Impact

Early disease detection can reduce treatment costs extensively because you are likely to catch the problems when they are easy and cheap to fix. The system **prevents major outbreaks that can cost thousands of dollars** in lost animals and reduced production, especially the conditions of modern farming where a lot of farm animals are reared in close proximity.

Use Case: Smart Money Planner

The Problem

Crop prices change every day. Fertilizer costs go up and down. Weather affects everything. It is hard to know when to sell your grain or buy supplies to make the most profit.

How AI Helps

The AI system looks at **crop prices, weather forecasts, and costs for seeds and fertilizer** all at the same time. It finds patterns that would take you hours to figure out by yourself.

The system creates simple reports that tell you things like "sell your corn next week because prices will probably drop" or "buy fertilizer now because costs are going up." It explains why it thinks this will happen using information you can understand.

Business Impact

Farms using smart planning systems typically can make more profit by buying and selling at better times, in other words, **better match the demand and supply**, taking into consideration all other factors like gestation period, etc. For example, a grain farmer can made an extra money per bushel by selling corn when the AI system said prices were at their peak, earning a few thousand dollars more, depending on quantity, than if he had sold at the usual time.

Use Case: Smart Food Waste Reduction and Inventory Management

The Problem

Restaurants throw away huge amounts of food every day because they prepare too much or ingredients spoil before being used. Food waste costs restaurants thousands of dollars each month and happens because it is hard to predict exactly how much food customers will order. Traditional inventory management relies on guesswork, leading to over-ordering ingredients that expire or running out of popular items during busy periods.

How AI Helps

AI systems track your food usage in real time by using **smart cameras** and **sensors**. The technology watches how much food gets thrown away, what ingredients are used most often, and which menu items sell best on different days. Based on this data, AI predicts exactly how much of each ingredient you need to order and when to order it.

The system can **automatically create purchase orders** when you are running low on specific items and warns you when ingredients are about to expire. Generative AI creates **easy-to-understand reports** that explain why certain foods are being wasted and suggests menu changes or portion adjustments to reduce waste.

Business Impact

AI-powered inventory tools can **optimize inventory levels**, especially perishables, as they monitor stock levels in real time, alerting staff to slow-moving or soon-to-expire ingredients. Restaurants using these systems typically can **reduce food waste** and **cut food costs**, while letting restaurant staff to focus more on providing good services and food.. The technology can pay for itself in the long run through reduced waste and better inventory management.

Use Case: Intelligent Customer Service and Order Management

The Problem

Taking orders, answering customer questions, and managing reservations takes up a lot of staff time, especially during busy periods. Customers often have to wait on hold or in line to place orders, which hurts their experience. Staff members spend time on repetitive tasks instead of focusing on food preparation and customer service that requires human attention.

How AI Helps

AI chatbots and voice assistants can take customer orders through your website, mobile app, or phone system. These systems understand natural language, so customers can say "I want a large pizza with pepperoni" and the AI will process the order correctly. The technology can also answer common questions about menu items, ingredients, and restaurant hours.

Generative AI creates **personalized responses** for customers and can suggest menu items based on their previous orders or dietary preferences. The system can also generate automated messages to update customers about their order status or delivery times.

Business Impact

Nearly all operators utilize online ordering, and AI systems can handle around most of routine customer interactions without human help, as they usually work within standard SOPs. This allows staff to focus on food quality and in-person customer service. Restaurants typically see **faster order processing, fewer mistakes, and higher customer satisfaction scores**.

Use Case: Dynamic Menu Optimization and Pricing

The Problem

Deciding which menu items to feature, how to price them, and when to run promotions requires analyzing lots of data about customer preferences, ingredient costs, and profit margins. Most restaurants make these decisions based on gut feeling rather than actual data, which can lead to promoting unprofitable items or pricing popular dishes too low.

How AI Helps

AI analyzes your sales data, ingredient costs, and customer behavior to **recommend which menu items to promote** and **how to price them**. The system can identify your most profitable dishes and suggest featuring them more prominently. It can also recommend creating daily specials using ingredients that need to be used soon.

Generative AI **creates compelling descriptions** for menu items and generates promotional content for social media or email marketing. The system can write engaging descriptions that highlight the most appealing aspects of each dish and create special offers that encourage customers to try new items.

Business Impact

AI-powered dynamic pricing tools adjust menu prices in real time to help restaurants move inventory more efficiently and reduce food waste. This data-backed approach empowers restaurants to build menus that resonate with customers and the bottom line while reducing waste behind the scenes. Restaurants using menu optimization typically **increase profit margins**, see **better inventory turnover**, and **reduce food wastage**.

Use Case: Creative Menu and Marketing Design with Generative AI

The Problem

Creating attractive menus, promotional posters, and marketing materials requires design skills and software that many restaurant owners do not have. Hiring graphic designers is expensive, and making changes to menus or promotions often involves waiting days and paying additional fees. This makes it hard for small restaurants to keep their marketing materials fresh and appealing.

How AI Helps

Generative AI design tools let you create professional-looking menus and posters by simply describing what you want. You can type instructions like "create a colorful menu for Italian restaurant with pizza and pasta photos" and the AI generates multiple design options. These tools include thousands of customizable templates and can automatically adjust colors, fonts, and layouts to match your restaurant's style.

Create stunning restaurant menus with thousands of customizable menu templates using A.I. technology. Free menu maker to design menu smarter, easier and faster. Tools like Designs.ai, Visme, and Canva's Magic Design can create custom, on-brand, and attention-grabbing designs in seconds.

Business Impact

Restaurant owners can update their menus and promotional materials instantly without paying designer fees. This flexibility allows restaurants to run more frequent promotions, update seasonal menus quickly, and maintain fresh marketing materials that attract customers. The **cost savings on design work** typically pay for the AI tools within the first month.

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Digital Sector

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/asean-digital-sector>

Use Case: Gojek's AI-Powered Platform, Indonesia



How AI Helps

Gojek uses a sophisticated AI system for **promotion automation** and **decision-making**. The AI analyzes user behavior, location data, and historical trends to provide personalized promotions and optimize the matching of users with drivers and merchants. The company has a clear governance structure for its AI, with a dedicated data science team and regular monitoring to ensure performance and fairness.



The Problem

As a super app, Gojek has to manage millions of daily transactions, from ride-hailing and food delivery to digital payments. Efficiently allocating promotions, drivers, and services to meet real-time user demand is a massive logistical challenge.



Business Impact

By using AI to automate and personalize promotions, Gojek has seen a significant increase in user engagement and retention. The optimized matching system has also improved service efficiency, leading to faster service times and a better user experience.

Source: [ASEAN Business Advisory Council](#)

Use Case: TIM (Telecom Italia)'s AI-Driven Network & Service Operations



How AI Helps

Google Cloud AI and **BigQuery ML** power predictive-maintenance models that analyse real-time network telemetry, automatically spotting anomalies before they impact customers. **Vertex AI** supports demand-forecasting and capacity-planning models, helping engineers allocate bandwidth where traffic surges are predicted. AI chatbots built with **Dialogflow** streamline customer support, handling common requests without human agents and routing complex issues intelligently.



The Problem

TIM needed to manage a massive telecom network, improve quality of service, and deliver new digital offerings while lowering operational costs. Legacy monitoring tools produced large volumes of raw data but lacked predictive capabilities, making it hard to prevent outages or scale new services efficiently.



Business Impact

AI-driven monitoring has reduced unplanned outages and improved network reliability, cutting both downtime penalties and support costs. Automated forecasting lets TIM deploy capacity more efficiently, saving on capital expenditure. AI chatbots shorten call-centre queues and improve customer satisfaction, enabling staff to focus on higher-value service issues.

Source: [Google Cloud Customer Story](#)

Use Case: Telefónica (Global ICT & Digital Services) – AI for Security and Customer Analytics



How AI Helps

Using **Google Chronicle Security Operations** with built-in machine-learning threat detection, Telefónica's SOC (Security Operations Centre) can identify and prioritise suspicious activity in near real time. Customer data from multiple business units flows into **BigQuery** and is modelled with **Vertex AI** to predict churn, personalise offers, and optimise marketing campaigns. Natural-language processing via **Dialogflow CX** powers virtual assistants that handle routine support and sales interactions.



The Problem

Operating across Europe and Latin America, Telefónica faced escalating cyber-security threats and an explosion of customer data. Manual threat analysis and traditional BI tools couldn't keep pace with the speed or complexity of attacks and customer behaviour patterns.



Business Impact

AI-driven monitoring has reduced unplanned outages and improved network reliability, cutting both downtime penalties and support costs. Automated forecasting lets TIM deploy capacity more efficiently, saving on capital expenditure. AI chatbots shorten call-centre queues and improve customer satisfaction, enabling staff to focus on higher-value service issues.

Source: [Google Cloud Customer Story](#)

Use Case: Find Your Best Customers, Faster

The Problem

A sales team spends a significant amount of time and effort manually searching for potential clients. This process is inefficient, often leads to an unqualified list of prospects, and can result in missed opportunities with high-value leads.

How AI Helps

AI-powered tools can **automatically scan vast amounts of public data**, including company websites, news articles, and social media posts, to identify businesses that match a company's ideal customer profile. It can detect key signals like a recent funding round, a new office opening, or a job posting for a specific role, indicating a potential need for a product or service.

Business Impact

By providing the sales team with a pre-qualified and prioritized list of leads, AI dramatically **improves sales efficiency and the quality of the sales pipeline**. The sales team can focus their energy on engaging with prospects who are most likely to convert.

Use Case: Predicting the Future of Your Business

The Problem

Traditional financial forecasting relies heavily on past performance and spreadsheets, making it slow to update and often inaccurate when unexpected changes occur in the market. This makes it difficult for business owners to make confident decisions about hiring, investments, or managing cash flow.

How AI Helps

Generative AI and machine learning can analyze not just a company's financial data, but also external factors like market trends, economic indicators, and even social media sentiment. It can then generate more accurate and flexible financial forecasts, budgets, and what-if scenarios, helping a business owner see a clearer picture of potential future outcomes.

Business Impact

This gives business owners a powerful tool for strategic planning. By having a more precise and up-to-date view of their financial future, they can make **better, faster decisions** that lead to stronger growth and more stable operations.

4

E-commerce

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/asean-e-commerce/>

Use Case: Shopee's Personalized Recommendation Engine, Singapore



The Problem

E-commerce platforms like Shopee have millions of products, making it difficult for users to find what they're looking for and for sellers to get their products seen.

How AI Helps

Shopee uses a sophisticated **AI-powered recommendation engine**. The system analyzes a user's browsing history, purchase behavior, and interactions with products to recommend relevant items. It also uses AI to optimize search results and product placement, ensuring users are shown the most relevant products.

Business Impact

This AI has been a key driver of sales and user engagement. By personalizing the shopping experience, Shopee has seen a significant increase in its conversion rate and a boost in overall transaction volume.

Source: [e27](#)

Use Case: Stitch Fix – AI for Fashion Design and Inventory

The logo for Stitch Fix, featuring the words "STITCH" and "FIX" in a bold, teal, sans-serif font. The letters are slightly spaced out, and the "I" in "STITCH" has a unique design with a horizontal bar.

How AI Helps

Data scientists use **Bayesian inference models**, **natural-language processing**, and **computer vision** to analyse style notes, fit feedback, and millions of product images. Algorithms recommend items for each client and even propose entirely new clothing designs by identifying emerging patterns in fabric, colour, and silhouette.



The Problem

Stitch Fix's subscription clothing service must predict customer style preferences and stock the right inventory. Traditional forecasting couldn't keep pace with fast-changing fashion trends or the need to personalise at the individual level.



Business Impact

Roughly 80% of customer purchases are influenced by AI-driven recommendations, helping Stitch Fix maintain high satisfaction and low return rates. AI-guided design also shortens the concept-to-market cycle, reducing unsold inventory and improving margins.

Source: [Stitch Fix](#)

Use Case: JD.com – Smart Logistics and Demand Forecasting



How AI Helps

JD.com employs machine-learning **forecasting models** that combine historical sales data, weather information, and real-time browsing behaviour to predict demand by region and SKU. In warehouses, **computer-vision** robots identify and sort packages, while AI-driven **route-optimisation** algorithms dynamically adjust last-mile delivery schedules.

The Problem

JD.com operates one of the world's largest e-commerce logistics networks in China. Meeting same-day or next-day delivery promises requires precise demand forecasts and efficient warehouse operations. Manual planning struggled with regional demand spikes and seasonal peaks.

Business Impact

The company reports forecast accuracy improvements, enabling leaner inventories and lower logistics costs. Automated warehouses reduce labour needs and increase throughput, allowing JD.com to maintain industry-leading delivery speeds and high customer satisfaction even during major shopping festivals.

Source: [Just One API](#)

Use Case: L'Oréal – AI-Powered Skin Diagnostics & Virtual Try-Ons

L'ORÉAL

The Problem

Shoppers often struggle to find the right shade of makeup or the best skincare routine for their individual complexion. In-store testers can be unhygienic or unavailable, and online purchases carry a high risk of mismatch, leading to product returns and customer frustration.

Source: [Bernard, Forbes](#)

How AI Helps

L'Oréal integrates **computer-vision** and **deep-learning models** into mobile apps and in-store kiosks. Customers upload a selfie or use a live camera feed; the AI analyses skin tone, undertone, texture, and concerns such as dryness, acne, or pigmentation. For makeup, **augmented-reality** technology overlays products—lipstick, foundation, hair colour—onto the live image so users can virtually “try on” different shades in real time. For skincare, the system generates personalised routines and product recommendations based on the detected skin condition.

Business Impact

These AI tools have transformed the buying experience by making it interactive and highly personalised. Customers gain confidence that recommended shades and treatments will match their real complexion, increasing conversion rates and reducing returns. L'Oréal reports higher engagement times in apps and on e-commerce platforms, stronger brand loyalty, and a measurable uplift in online sales, all while eliminating the hygiene concerns and logistical costs of physical testers.

Use Case: Smart Product Recommendations

The Problem

Customers cannot find products they want to buy on your website. They see too many choices or the wrong items, so they leave without purchasing anything. This means lost sales and lower profits.

How AI Helps

AI watches what each customer looks at and buys, then shows them products they are most likely to purchase. If someone buys outdoor gear, the system automatically shows them camping equipment and hiking boots. The AI learns each customer's preferences and creates a **personalized shopping experience**.

Business Impact

Studies show personalized recommendations can **improve sales** significantly as compared to standardized recommendation rules. Customers find what they want faster, buy more items, and return to shop again. Smarter recommendation can **increase the customer lifetime value and customer loyalty** at the same time.

Use Case: Smart Inventory Management

The Problem

You order too much of products that do not sell while running out of popular items. This wastes money on storage and disappoints customers when they cannot buy what they want.

How AI Helps

AI analyzes your sales history, seasonal patterns, and market trends to predict exactly how much of each product you will need. The system automatically suggests **when to reorder items** and warns when you might run out of popular products.

Business Impact

AI typically **reduces inventory costs** while keeping popular items in stock. Companies like Target use AI forecasting to manage inventory better. You waste less money on unsold products and keep customers happy with better product availability.

Use Case: Smart Pricing

The Problem

Setting the right prices for hundreds or thousands of products is difficult when competitor prices and customer demand change constantly. Wrong pricing means lost sales or reduced profits.

How AI Helps

AI monitors competitor prices, customer behavior, and your inventory levels to automatically adjust prices for maximum profit. When demand is high, prices can increase slightly. When you have too much inventory, prices can drop to move products quickly.

Business Impact

Companies like Amazon use AI to adjust millions of prices daily based on market conditions. This strategy helps **maximize revenue** while staying competitive. You make more profit from each sale while responding quickly to market changes. Using AI it allows the retailer to tap onto revenue potential further of an item.

Use Case: AI Customer Service

The Problem

Customers expect immediate answers about products, orders, and returns. Traditional customer service keeps people waiting and cannot provide 24-hour support. This leads to lost sales and frustrated customers.

How AI Helps

AI chatbots answer customer questions **instantly** about product details, order status, and return policies. The system handles multiple customers at once and works **24 hours** a day. For complex issues, the AI transfers customers to human staff.

Business Impact

AI handles most of routine customer questions without human help because there is a standard operating procedures to them. This reduces labor costs while improving response times. Customers get immediate answers, leading to more completed purchases and **higher satisfaction**.

5

Tourism

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/asean-tourism-sector>

Use Case: Changi Airport's AI-powered Passenger Flow Management and Security Check, Singapore



The Problem

Managing the flow of millions of passengers through a large hub like Changi Airport is a major logistical challenge. Bottlenecks at security checkpoints, immigration, and boarding gates can lead to delays and a poor passenger experience.

At busy checkpoints, manually screening X-ray images of carry-on baggage is time-consuming and labor-intensive. The process slows down throughput and increases wait times for travelers, especially during peak hours. Human screeners might miss subtle or concealed prohibited items, and the need for consistent accuracy can strain resources.

Source: [Changi Airport Group](#)

How AI Helps

Changi Airport uses an AI system to analyze real-time data from sensors and cameras to monitor passenger flow. The AI predicts potential bottlenecks and recommends adjustments, such as opening more check-in counters or re-allocating staff to a different security gate.

An AI/ML-powered system—referred to as APIDS—that automatically analyzes X-ray images of carry-on luggage to flag prohibited items was implemented.

Business Impact

The AI system has improved operational efficiency by optimizing resource allocation and has reduced wait times for passengers. This contributes to Changi's reputation as a world-class airport and enhances the overall tourism experience in Singapore.

Changi Airport anticipates security checks could become up to 50% quicker thanks to the AI system.

Use Case: Expedia's Role in Advancing Quality Tourism in Southeast Asia



The Problem

Southeast Asia's tourism sector has rebounded strongly post-pandemic, yet challenges remain in ensuring sustainable, inclusive, and high-quality tourism growth. Popular destinations suffer from overtourism, while lesser-known areas struggle to attract visitors. Environmental degradation, inconsistent visitor experiences, and lack of accessibility—particularly for persons with disabilities—further hinder balanced development. Merely increasing tourist arrivals is no longer sufficient; the focus must shift toward quality tourism that delivers economic value while preserving culture, promoting inclusivity, and minimizing environmental impact.

Source: [Expedia Group](#)

How AI Helps

Expedia Group leverages artificial intelligence and big data to support the shift toward quality tourism in the region. Its platform analyzes vast volumes of traveler data to **forecast demand trends, promote lesser-known destinations, and match travelers with eco-conscious or accessible accommodations.** Features like sustainability filters, accessibility guides, and personalized travel suggestions help surface relevant options aligned with traveler values. AI also enables real-time targeting of high-value, last-minute, or sustainability-oriented tourists, empowering tourism boards and partners to design smarter campaigns, reduce overcrowding in hotspots, and encourage more equitable tourism distribution across the region.

Business Impact

The use of AI-driven insights has contributed to significant tourism recovery in ASEAN, with countries like Vietnam seeing a 40% rise in visitor numbers. Popularity rankings have improved, with Thailand becoming the most visited city globally and Indonesia climbing 10 spots in the WEF tourism index. More importantly, Expedia's tools have supported the growth of sustainable travel habits—90% of travelers now seek eco-friendly options—and encouraged dispersal of visitors to Tier-2/3 cities.

Use Case: AirAsia – AI-Driven Dynamic Pricing and Demand Forecasting, Malaysia



The Problem

AirAsia operates hundreds of daily flights across Asia and faces intense competition and volatile passenger demand. Traditional revenue-management systems struggled to account for rapidly changing factors such as fuel prices, seasonal travel spikes, and sudden events (e.g., weather disruptions or regional holidays). This often resulted in sub-optimal seat pricing, inaccurate demand forecasts, and missed revenue opportunities.

Source: [Google Cloud's Customer Story](#)



How AI Helps

Using Google Cloud's **BigQuery** and **Vertex AI**, AirAsia integrates historical booking data, real-time search trends, competitor pricing, and external variables such as weather and local events. Machine-learning models **forecast demand for each route and time window**, then feed predictions into a dynamic pricing engine that automatically adjusts fares to balance load factors and revenue. **AI-powered analytics dashboards** give revenue managers instant insights, allowing them to fine-tune campaigns and capacity planning.



Business Impact

AI-driven forecasting has improved seat-yield management and boosted average revenue per flight while reducing last-minute unsold inventory. More accurate predictions of passenger loads help optimize crew scheduling and fuel planning, lowering operational costs and emissions. The ability to respond in near real time to market changes strengthens AirAsia's competitiveness and enhances the booking experience for customers through more consistent availability and fair, demand-aligned pricing.

Use Case: AI-Driven Predictive Maintenance for Urban Public Transport

The Problem

A metropolitan transit authority operated thousands of buses and trains but relied on fixed-interval maintenance schedules. Unexpected mechanical failures caused service delays, costly emergency repairs, and passenger complaints.

How AI Helps

Machine-learning models ingest sensor readings from engines, braking systems, and climate controls, along with historical maintenance logs and environmental data. The system **predicts component failures weeks in advance** and **schedules service proactively**. **Computer-vision cameras** at depots automatically flag visual wear or damage during daily inspections.

Business Impact

Unplanned breakdowns fell, cutting overtime and spare-parts costs while improving on-time performance. Riders benefit from more reliable service, and the agency meets stricter safety and emissions regulations without adding maintenance staff.

Use Case: AI-Enhanced Customer Support for a Global Hospitality Chain

The Problem

A large hospitality group struggled with rising call-centre volumes and inconsistent service quality across regions. Guests often faced long wait times, and agents spent excessive time searching multiple knowledge bases for answers, leading to higher costs and lower satisfaction scores.

How AI Helps

A cloud-based generative AI platform **unifies reservation data, loyalty-program details, and property-specific information**. Natural-language models analyse incoming messages in real time, recommend personalised responses to agents, and automatically translate conversations across languages. Predictive analytics forecast call spikes and route queries to the best-suited agents or self-service bots.

Business Impact

Average handling time dropped, while first-contact resolution rates improved significantly. Customers experience faster, more accurate support, and the company reduces staffing costs while freeing human agents to handle complex, high-value interactions.

6

Science and Technology

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/asean-science-technology-and-innovation>

Use Case: Grab's AI for Safety and Fraud Detection, Singapore



The Problem

As a platform with millions of users and drivers, Grab faces the constant challenge of ensuring passenger safety and preventing fraudulent activities.

Source: [Grab](#)

How AI Helps

Grab uses AI and machine learning to monitor real-time data and detect unusual patterns that might indicate a safety risk or fraudulent activity. The AI system can identify potential issues, such as a driver deviating from the planned route or a payment pattern that suggests fraud, and flag them for human intervention. This includes **facial recognition** for driver verification and **sentiment analysis** of in-app messages to detect distress.

Business Impact

The AI has significantly reduced the rate of fraudulent transactions and safety incidents, building greater trust among users and drivers. This has a direct positive impact on Grab's brand reputation and business sustainability.

Use Case: University Research Teams Using Google's AI Co-Scientist



The Problem

Researchers in biomedical and physical sciences often spend weeks combing through thousands of journal articles to stay current, spot gaps in knowledge, and design new experiments. This heavy literature review delays discovery, consumes grant funding, and limits the time scientists can devote to hands-on experimentation.

Source: [Google](#)



How AI Helps

University labs now work with Google's **AI Co-Scientist**, a **Gemini-powered multi-agent system** that acts as a virtual research partner. Scientists feed it research questions or datasets; the AI rapidly scans global scientific literature, highlights contradictory findings, suggests under-explored connections, and drafts experimental designs. Instead of replacing researchers, it gives them a constantly updated map of relevant work and candidate hypotheses, freeing them to focus on critical thinking and lab work.



Business Impact

Teams report that literature-review time has dropped from weeks to days, enabling them to launch experiments sooner and submit grant proposals ahead of competitors. Faster hypothesis generation and better experiment planning reduce wasted materials and improve the odds of publishable results, helping labs stretch budgets and accelerate breakthroughs.

Use Case: Clinicians and Medical Researchers Using Vertex AI Search for Healthcare



Vertex AI

The Problem

Doctors, nurses, and clinical researchers need to find specific details buried in electronic medical records—lab results, imaging notes, prior medications—spread across multiple systems and formats. Traditional keyword searches are slow and often miss crucial context, delaying diagnoses and research studies.

Source: [Google](#)



How AI Helps

Hospitals and research institutions deploy **Vertex AI Search** for Healthcare, which lets clinicians type natural-language questions such as “patients with elevated liver enzymes and prior statin use in the past three months.” The system semantically searches structured data and unstructured notes, summarising key findings and surfacing relevant records within seconds. Physicians stay in the loop, reviewing AI-generated summaries before making clinical decisions.



Business Impact

Clinicians spend less time digging through records and more time with patients, leading to faster, more accurate treatment plans. Research teams can identify patient cohorts for clinical studies in hours instead of weeks, accelerating trial recruitment and new-drug investigations. Hospitals benefit from improved care quality, reduced administrative costs, and higher staff satisfaction.

Use Case: Real-Time Coral Reef Monitoring

The Problem

Marine biologists working to protect coral reefs struggle to track bleaching across thousands of square kilometres. Satellite images often arrive days late, and manual dive surveys are costly and limited in coverage, making it nearly impossible to spot early stress signals or deploy conservation measures before serious damage occurs.

How AI Helps

Autonomous underwater drones equipped with hyperspectral cameras stream live video to a cloud-based AI platform. **Computer-vision models detect subtle colour changes and microscopic organism patterns** that signal early bleaching. Machine-learning algorithms combine these observations with real-time sea-surface temperature and nutrient data to predict where bleaching is likely to spread next.

Business Impact

Researchers can now identify vulnerable reef zones within hours rather than weeks, directing divers and restoration crews to the highest-priority sites. Early intervention improves coral survival rates and trims research costs by an estimated 40 percent through fewer unnecessary survey trips. Coastal communities and tourism operators benefit from healthier reefs and more reliable ecosystem services.

Use Case: Accelerating New Material Discovery

The Problem

A private research team exploring superconducting materials faces experimental cycles that can last for years. Testing every potential alloy combination in the lab is expensive, and the number of possible compounds is virtually limitless, slowing the path to breakthrough discoveries.

How AI Helps

A generative chemistry model proposes novel crystal structures likely to exhibit high-temperature superconductivity. Reinforcement-learning algorithms rank candidates by predicted stability and manufacturability, while robotic labs automatically synthesise and test the most promising formulas. Results feed back into the model, continually improving predictions and shrinking the experimental search space.

Business Impact

This closed AI–robotics loop cuts the time to identify viable material candidates from roughly five years to under eighteen months. Faster breakthroughs strengthen intellectual-property portfolios, **attract new funding**, and **enable earlier commercial partnerships** for next-generation electronics. Around-the-clock automated experimentation reduces human error and makes more efficient use of costly laboratory resources.

6

Transport

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/transport/>

Use Case: Tuas Port's AI-driven Logistic Hub, Singapore



How AI Helps

Tuas Port is a fully automated, AI-driven logistics hub. It uses AI to orchestrate real-time operations, including **autonomous guided vehicles (AGVs)** and **AI-enabled cranes**. The AI system plans container movements, optimizes vessel loading and unloading, and manages the flow of trucks in and out of the port.



The Problem

Traditional ports are labor-intensive, space-constrained, and can be inefficient due to manual processes and human error, leading to slower turnaround times and higher costs.



Business Impact

Since launching operations in 2022, the port has handled over 10 million TEUs (Twenty-foot Equivalent Units) with zero on-site operators, dramatically increasing efficiency and reducing operational costs. This has transformed the port into a model of digital competitiveness.

Source: [Vu Minh Khuong, LKYSPP, NUS](#)

Use Case: Yamato Transport — Last-Mile Delivery Optimization, Japan



The Problem

Yamato Transport has faced rising volume of package deliveries (especially B2C), uneven workloads for drivers, increasing delays, pressure from labor shortages (aging workforce, fewer drivers), and the need to respond to climate change and rising customer expectations. Traditional routing and operations relied heavily on driver experience and manual planning, which did not scale well with fluctuations in orders or constraints (time, traffic, driver hours).

Source: [Google Cloud's Customer Story](#)



How AI Helps

Yamato Transport combines **Google Maps geospatial data** with machine-learning algorithms running on Google Cloud to optimise last-mile delivery. The AI models ingest historical delivery records, live traffic patterns, package volumes, and driver availability to predict demand spikes and travel times for every route. Using these predictions, the system automatically generates the most efficient delivery schedules, dynamically reallocates territories among drivers, and recalculates optimal paths in real time as new orders or traffic conditions emerge. By incorporating drivers' local knowledge—such as preferred parking spots or building access codes—into the training data, the AI continuously learns and refines its routing decisions, improving accuracy and efficiency with each delivery cycle.



Business Impact

By deploying AI-driven route optimisation and dynamic scheduling, Yamato Transport has increased delivery efficiency and reduced costs by cutting total route distance, fuel consumption, and vehicle wear, while balancing driver workloads to ease labour shortages and prevent burnout. More accurate demand forecasting and real-time re-routing have improved on-time arrivals and customer satisfaction, and the shorter routes and reduced idling have lowered CO₂ emissions, enabling Yamato to handle rising e-commerce volumes without adding proportional staff or fleet capacity, thereby strengthening both profitability and sustainability.

Use Case: Penske Truck Leasing — Predictive Maintenance & Fleet Efficiency



The Problem

Penske, which maintains and leases a fleet of ~433,000 trucks, found itself constrained by traditional maintenance strategies. These were mostly mileage-based or reactive (fixing failures when they occur), which led to unexpected breakdowns, high downtime, unplanned repair costs, and inefficiencies in utilization.

Source: [Business Insider](#)



How AI Helps

Penske implemented an AI system called Fleet Insight / Catalyst AI that ingests telematics data from trucks (performance metrics, sensor data, etc.), processes around 300 million messages daily across the fleet, and applies algorithms to **detect patterns that indicate emerging mechanical issues** (e.g. wear and tear in hose, tires). The AI flags these before they become failures, so maintenance can be scheduled proactively.



Business Impact

This predictive approach has reduced unplanned downtime, leading to cost savings (fewer emergency repairs, better parts management), improved reliability for clients leasing trucks (who experience fewer disruptions), better fuel efficiency (because vehicles are maintained in better condition), and improved vehicle availability. For example, some clients (like Darigold, Honeyville) report that this helps them benchmark performance vs. industry standards, understand under-performance, and rectify it. Overall, Penske is enhancing customer satisfaction and gaining competitive advantage through more dependable service and lower total cost of ownership for fleet users.

Use Case: DHL – Route Planning & Last-Mile Delivery Optimization



How AI Helps

DHL has applied AI in its route-planning systems. The AI ingests historical delivery data, traffic patterns, weather, and real-time updates, and then recommends optimized delivery routes. These systems can **dynamically adjust route plans** based on live inputs. In addition, they integrate delivery constraints (vehicle size, delivery windows, order sequence) and “courier knowledge” (e.g. experience of drivers about local parking or access) to refine optimization. They also run what-if simulations to anticipate demand surges or delays.



The Problem

DHL faces typical logistics constraints: last-mile delivery is costly and inefficient due to variable traffic, changing delivery volumes, and customer expectations for precise delivery windows. Also, sub-optimal routing leads to excessive fuel consumption and longer delivery times. Dispatchers and traditional planning tools struggle to adjust rapidly in response to real-time changes (traffic, order spikes, weather, etc.).



Business Impact

Through AI-powered route optimization, DHL achieves faster deliveries, more accurate estimated delivery times (improving customer satisfaction), lower fuel consumption and emissions, and less wasted time from drivers being delayed or idle. Efficiency gains reduce operating costs and contribute toward sustainability goals. Also, by having more predictable delivery windows and more reliable timing, customer trust improves, possibly reducing complaints or missed deliveries. DHL cites improvements in speed and reductions in fuel waste due to smarter routing.

Source: [DHL](#)

Use Case: Smart Warehouse Inventory Management

The Problem

Managing inventory in large warehouses is challenging with thousands of moving items. Workers spend hours searching for products, counting stock by hand, and updating systems. Errors—like misplaced items or inaccurate records—cause delays, extra costs, and lost sales when products cannot be found or popular items run out.

How AI Helps

AI systems use cameras and smart sensors to watch your warehouse **24 hours** a day. The technology can **automatically count items** on shelves, **read barcodes** even when they are partially covered, and track where everything is located in real time. When workers move products, the AI system updates the inventory numbers immediately without anyone typing anything into a computer. Generative AI creates **easy-to-understand reports** that explain what is happening in your warehouse. Instead of looking at complicated spreadsheets, you get simple summaries like "You are running low on Product A in Section 5" or "Product B has been sitting in the same spot for 30 days." The system can also generate restocking suggestions and predict which items you will need to order next week based on your sales patterns.

Business Impact

Companies using smart inventory systems typically **reduce their inventory costs** by increasing the turnover while improving order accuracy. Workers spend less time searching for items and more time on important tasks. One major benefit is that AI helps automate tasks, boost efficiency, and manage things like inventory, equipment, and deliveries more smoothly. The technology pays for itself quickly because it prevents costly mistakes like shipping wrong items to customers or running out of popular products. Many warehouses see improvements within the first month of installation.

Use Case: Intelligent Delivery Route Planning

The Problem

Planning delivery routes for multiple trucks and drivers is extremely complicated. You have to consider traffic, weather, customer time preferences, truck capacity, and driver schedules all at the same time. Traditional route planning often results in drivers getting stuck in traffic, making unnecessary stops, or running out of time to complete all deliveries. This wastes fuel, makes customers unhappy, and increases costs.

How AI Helps

AI route optimization uses machine learning and real-time data analytics to determine the **most efficient delivery routes dynamically**. The system looks at current traffic conditions, weather forecasts, and delivery priorities to create the best possible routes for each driver. It continuously adjusts routes throughout the day when unexpected problems arise, like traffic accidents or urgent delivery requests.

Generative AI creates clear, step-by-step instructions for drivers that are easy to follow. Instead of complex route maps, drivers get simple directions with explanations like "Take Highway 5 to avoid construction on Main Street" or "Deliver to Customer A first because they close at 3PM." The system can also generate customer communication messages that automatically update people about delivery times.

Business Impact

AI route optimization generates the most time-saving routes, minimizing the distance traveled, in turn **reduce carbon footprint** and **fuel consumption** while **boosting on-time deliveries**. Companies can look to **reduction in fuel costs** and can **complete more deliveries** with the same number of drivers. The system helps logistics companies respond effectively and swiftly to last-minute delivery requests while maintaining efficiency. This improved service makes customers happier and helps companies win more business from competitors.

Use Case: Predictive Equipment Maintenance

The Problem

Logistics companies depend on trucks, conveyor belts, forklifts, and other equipment that can break down unexpectedly. When a delivery truck breaks down during a route, it can delay dozens of deliveries and cost thousands of dollars in overtime and customer complaints. Traditional maintenance schedules service equipment at fixed times, whether it needs it or not, which wastes money on unnecessary repairs while still allowing unexpected breakdowns.

How AI Helps

AI systems monitor the health of your equipment by analyzing data from sensors that track things like engine temperature, vibration patterns, and usage hours. The technology learns what normal operation looks like for each piece of equipment and can **predict when something is likely to break down** days or weeks before it actually happens. Generative AI creates maintenance schedules and work orders that make sense for your business. Instead of generic maintenance reminders, you get specific recommendations like "Truck 7 needs brake inspection next Tuesday" or "Conveyor Belt 3 shows signs of bearing wear and should be serviced within 5 days." The system explains why each maintenance task is needed and what will happen if you delay it.

Business Impact

Predictive maintenance typically **reduces equipment downtime and cuts maintenance costs**, while allowing equipments continue to be performing well. Companies avoid expensive emergency repairs and can plan maintenance during slow periods instead of having equipment fail during busy times. From smart machines that track stock to systems that predict when equipment needs fixing, AI is becoming a key tool in modern logistics operations. The technology helps companies maintain reliable service while controlling costs, leading to better customer satisfaction and higher profits.

7

Standard and Conformance

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/standard-and-conformance/>

Use Case: AI for Quality Control and Inspection at Flex, Malaysia



How AI Helps

Flex, a global manufacturing services company with a significant presence in Malaysia, has deployed **AI-powered computer vision systems for quality control** on its production lines. High-speed cameras capture images of products, and an AI model, trained on thousands of images of both perfect and defective items, can detect minuscule flaws with a high degree of accuracy. The system provides real-time feedback, allowing for immediate adjustments to the production process.



The Problem

In the manufacturing sector, ensuring that every product meets strict quality standards is a labor-intensive and error-prone process. Manual inspections often miss subtle defects, leading to product recalls and reputational damage. This lack of consistent quality across the supply chain creates non-tariff barriers, as importing countries may require additional checks to verify product safety and quality.



Business Impact

Flex, a global manufacturing services company with a significant presence in Malaysia, has deployed **AI-powered computer vision systems for quality control** on its production lines. High-speed cameras capture images of products, and an AI model, trained on thousands of images of both perfect and defective items, can detect minuscule flaws with a high degree of accuracy. The system provides real-time feedback, allowing for immediate adjustments to the production process.

Source: [Flex](#)

Use Case: Network Rail — Safety-Critical Communication (SCC) Auditing, UK



How AI Helps

Network Rail, together with Capgemini, built a proof-of-concept AI system that **listens to or analyses call recordings (audio), transcribes them, and evaluates whether key criteria of the SCC standard are met** (clarity, brevity, professionalism). The system provides automated scoring or flagging of calls that fail to meet standards, so that management can focus on those for review or remedial training.



The Problem

Network Rail needed to ensure that verbal communications between its various engineers, signallers, and track works teams comply with its Safety-Critical Communication (SCC) standards. SCC rules require calls to be accurate, clear, professional, and brief. With thousands of calls daily, manual audits are costly, inconsistent, and slow, making it hard to know which employees or calls are non-compliant and to improve training or processes accordingly.

Source: [Capgemini](#)



Business Impact

AI-auditing has enabled more frequent and scalable monitoring of SCC compliance. Instead of auditing only a small sample of calls, Network Rail can now cover a much larger fraction, identifying training needs more precisely and earlier. This improves safety, because communication errors are less likely, and helps reduce risks associated with miscommunication in safety-critical contexts. Also, consistency in evaluation is higher, and resources (auditors' time) are used more efficiently. Over time, this leads to better compliance rates, fewer incidents due to misunderstanding, and lower costs tied to accidents or miscommunication.

Use Case: AES – Scaling Safety / Health Audits with GenAI Agents



How AI Helps

AES implemented generative AI agents built on Google Cloud's **Vertex** AI and Anthropic's **Claude** models to streamline its safety and compliance audits. **These agents automatically review and summarise hundreds of pages of multilingual documentation, extract key compliance requirements, and highlight potential gaps or issues for auditors to verify.** By handling the time-consuming tasks of document analysis, cross-checking, and translation, the AI enables human auditors to focus on higher-value judgement and decision-making. The system operates at scale across AES facilities, allowing audits to be launched and completed simultaneously across multiple sites and continually refined as new regulatory standards or internal policies are introduced.



Business Impact

By integrating generative AI into its audit process, AES cut the average audit time from roughly two weeks to about one hour and lowered the cost per audit by an estimated 99 percent. This dramatic efficiency gain allows the company to double the number of safety and compliance audits it performs each year without adding staff, while the AI's consistent analysis improves accuracy by about 20 percent. Faster, more reliable audits mean issues are identified and resolved sooner, strengthening regulatory compliance, reducing operational risk, and enhancing the overall safety culture across all AES facilities.

The Problem

AES, a large energy company with operations across many plants and power-generation facilities, conducts over 1,500 safety and health management program audits per year. These audits include both internal assessments and external audits of sites. Each audit is very labour-intensive, involving document reviews (sometimes hundreds of pages), checks for accuracy and thoroughness, plus manual compilation of audit reports. Traditionally, each audit took about 14 days and required ~100 employee hours per audit. This scale and time commitment limited how often audits could be done, delayed findings, and made it harder to ensure consistent compliance across all sites.

Source: [Google Cloud's Customer Story](#)

Use Case: Automotive Manufacturer — AI in Quality Management / Conformance Checking

The Problem

An automotive manufacturer had quality control workflows and supplier conformance checks that were heavily manual: inspections of parts, defect detections, supplier deviation reports. These processes were slow, error-prone, and costly, especially when scaling up production or dealing with many suppliers. Late detection of defects caused reworks or recalls.

Source: [Qualityze](#)

How AI Helps

They implemented an **AI-powered Quality Management System (QMS)** which ingests production line data, sensor readings, supplier data, and customer feedback. The system uses anomaly detection models to spot deviations from quality specifications, flags likely defects, and provides real-time monitoring of quality metrics. It also helps predict which components or suppliers might fail conformance so that preventive action can be taken.

Business Impact

After deploying the AI-augmented QMS, the manufacturer saw roughly 30% reduction in quality-related costs (fewer reworks, less waste), and about 20% increase in customer satisfaction because product defects were caught earlier and fewer faulty products shipped. The speed of inspections and conformance checking improved, allowing more frequent checks and better supplier oversight. Over time there are also improvements in supplier performance and reliability since preventive warnings allow better supplier management.

Use Case: Robotic Optimisation of Powdered Beverages (Food & Beverage)

The Problem

In beverage product development (e.g. powdered drinks like cappuccino), there are many variables (e.g. foam quality, powder composition, mixing conditions) that affect the final sensory attributes. Traditional experimentation is slow, laborious, and limited in exploring the full combination space.

How AI Helps

A robotic system is used to **automate the preparation of samples**. It uses computer vision to assess foam quality, and Bayesian optimization to explore the parameter space (different mixtures / preparation processes) more efficiently. The feedback (foam quality, etc.) loops back into the optimization. This enables exploring many more combinations in a structured way, with quicker iteration.

Business Impact

By speeding up experimentation, companies can more quickly identify optimal formulations (improving product quality) and reduce R&D costs. Also, repeatability is much higher than manual experiments, which helps in consistent quality in production and product launch speed. (The paper doesn't give full commercial metrics like revenue uplift, but the impact in a lab/R&D context is strong.)

Source: [arxiv](#)

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Services

For further insights and more comprehensive information on the sector across ASEAN, please refer to the following link:
<https://asean.org/our-communities/economic-community/services/>

Use Case: AIA's AI-Powered Customer Service Chatbot, Singapore



How AI Helps

AIA, a major insurance provider in Asia, has deployed an AI-powered chatbot to handle customer inquiries. The chatbot uses natural language processing (NLP) to understand customer questions and provide instant, accurate responses. It can handle a wide range of tasks, from checking policy details to initiating claims, and only escalates to a human agent when necessary.



The Problem

The insurance industry relies heavily on customer service, but traditional methods are often slow and inefficient. Customers may have to wait on hold to ask simple questions or process claims.



Business Impact

The chatbot has significantly reduced the burden on the human customer service team, allowing them to focus on more complex cases. It has also improved customer satisfaction by providing immediate support and has led to a reduction in operational costs.

Source: [Benjamin Ang, Asia Advisor Network](#)

Use Case: DBS and AI in Banking, Singapore



The Problem

DBS recognized that traditional banking models were becoming increasingly unsustainable in the digital age: many operational tasks were manual and inefficient (e.g., compliance, fraud detection, customer engagement), customer expectations for personalization and responsiveness were rising, and competition from digital-native players was intensifying. To stay relevant and competitive, DBS needed to transform into a digitally native organization capable of delivering faster, smarter, and more secure banking experiences while optimizing internal processes.

Source: [DBS](#)



How AI Helps

DBS has embedded AI and machine learning deeply across both front-end and back-end operations by deploying over 800 AI models across more than 350 business use cases. These range from chatbots (e.g., Digibot) for customer service, algorithmic wealth and portfolio guidance with its digiPortfolio, dynamic fraud and AML detection through real-time transaction surveillance, hyper-personalized “nudges” to aid financial planning, to workforce intelligence like HR attrition modeling. DBS also developed governance frameworks and the “P-U-R-E” ethical AI model to ensure responsible deployment.



Business Impact

DBS’s AI industrialization has yielded significant business outcomes: the bank anticipates that the measured **economic impact from AI initiatives will exceed SGD 1 billion by 2025**, with sequential annual doubling of impact metrics. With hyper-personalized customer communications (45 million monthly communications/nudges) and proactive transaction monitoring, DBS enhances customer trust and precision targeting of products and services. The bank has become recognized globally—#1 in AI strategy leadership among financial institutions in the Global Evident AI Index, and winner of the Global Model Bank Award for AI Industrialization.

Use Case: Discover Financial Services — Generative AI for Customer Care



How AI Helps

Discover collaborated with Google Cloud to **integrate generative AI tools into their customer care workflow**. The AI assists agents by surfacing recommended responses, summarizing relevant content, guiding through knowledge base articles, and pulling in context automatically from past interactions. This reduces the time agents spend looking for information, allows more consistent responses, and enables more personalized engagement with customers. The tools are aimed at increasing agent productivity, supporting faster resolution, and reducing the cognitive load on agents.



The Problem

Discover Financial Services operates a large-scale customer care organization with ~10,000 agents. Their challenges included long resolution times, inconsistent customer responses, high agent load, and the need for more personalized, efficient support. The existing system required agents to manually search through knowledge bases, documents, and prior cases to find appropriate responses, slowing things down and sometimes leading to sub-optimal customer experiences.

Source: [Discover](#)



Business Impact

By bringing in generative AI, Discover has improved response times in its call centers (i.e. customers get answers faster), and enhanced the overall customer experience. Agents are able to handle queries more efficiently, which improves operational throughput. Additionally, personalized responses and more consistent service contribute to increased customer satisfaction. Though exact percentage improvements in resolution time or satisfaction weren't detailed in the public source, the scope (10,000 agents) suggests large scale efficiencies and likely cost savings via reduced average handling time.

Use Case: Answering Common Questions Instantly with a Chatbot

The Problem

Your customers often ask the same questions repeatedly: "What are your business hours?", "How do I track my order?", or "What is your return policy?". Answering these takes up a lot of time for your staff, especially during busy periods. Customers get frustrated waiting for a reply, and your employees are stuck with repetitive work instead of focusing on more important tasks.

How AI Helps

You can use a simple, low-cost AI chatbot on your website or social media messaging app. You teach the chatbot by giving it the answers to your most frequent questions. When a customer asks one of these questions, the chatbot will immediately provide the correct answer. If the question is too complex, the chatbot can be set up to politely tell the customer it will connect them with a human staff member during business hours.

Business Impact

Using AI support tools saves money by reducing the need to hire extra staff for basic questions. Customers enjoy instant answers anytime—even at night or on holidays—which makes them feel valued and cared for. At the same time, employees benefit from **improved morale** as they are freed from repetitive tasks and can focus on solving more meaningful, complex issues that require a human touch.

Use Case: Sorting and Prioritizing Customer Emails

The Problem

Your customer service inbox is a mess. Emails pile up, and it's hard to tell which ones are urgent complaints and which are simple questions. Important emails might get buried and go unanswered for too long, leading to angry customers. Your staff has to spend time just reading and sorting all the emails before they can even start to help.

How AI Helps

An AI-powered email system can read every incoming email and automatically understand its main topic. For example, it can see that an email is about a late delivery, a broken product, or a request for a refund. It can then automatically tag the email and put it into the right folder for the right person to handle. It can even prioritize urgent issues, putting them at the top of the list for your team to see first.

Business Impact

AI-enabled systems deliver faster service by spotting and resolving urgent problems quickly before they escalate. They also improve organization by removing the need for staff to manually sort through large volumes of emails, allowing the team to work more efficiently. At the same time, customers gain confidence in your business, knowing their concerns are addressed promptly, which **strengthens trust and long-term loyalty**.

Use Case: Creating a 24/7 Virtual Assistant on WhatsApp

The Problem

Many customers in ASEAN prefer to communicate using messaging apps like WhatsApp. However, your business can't have a staff member available 24/7 to answer messages, especially if your customers are in different time zones. This means you miss out on potential sales and customer service opportunities outside of your normal working hours.

How AI Helps

You can set up an AI virtual assistant on your business's WhatsApp account. This AI can be trained to answer common questions, provide product details, and even help customers place an order or check an order status. Since it's an AI, it can work around the clock, every day of the year.

Business Impact

The business impact is significant: you never miss a customer because your business is always “open” on the platforms where they are already active, allowing you to answer questions and even close sales anytime. You can also **reach customers across different countries and time zones without hiring extra staff**. Best of all, customers enjoy the convenience of communicating through messaging apps they already use daily, making the experience simple and familiar.

Use Case: Getting Customer Feedback Automatically

The Problem

After a customer has been helped by your team, you want to know if they were happy with the service. Manually calling or emailing every customer for feedback is slow and labor-intensive. It's difficult to get a good sense of what your customers truly think, and you can't easily spot repeating problems.

How AI Helps

After a customer service interaction (like a chat or an email), an AI can **automatically send a short message asking for feedback**. The AI can then read the customer's response and, using "sentiment analysis," understand if the customer's tone is positive, negative, or neutral. It can even highlight specific keywords that are mentioned often, like "slow" or "helpful," giving you a clear picture of what's working and what's not.

Business Impact

Customer feedback provides quick insights without extra effort, helps spot complaint patterns to address root problems, and empowers your team by recognizing good performance while guiding training needs—ultimately improving service and motivation.

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Manufacturing

Use Case: Siemens – Smart Factory Quality Control



How AI Helps

Siemens Energy uses Google Cloud **Vision AI** and **Vertex AI** to analyse high-resolution images from factory inspection stations. Computer-vision models trained on historical defect data automatically flag surface cracks or irregularities in real time, while **BigQuery** stores and correlates results with upstream process data to trace root causes.



The Problem

Gas-turbine production involves thousands of precision parts where even minute surface defects can cause costly failures. Manual inspections were slow, labour-intensive, and sometimes missed sub-millimetre flaws, leading to rework, warranty claims, and production delays.



Business Impact

Automated inspection reduces detection time from hours to seconds, cutting defect-related rework by an estimated 30%. Early detection lowers warranty costs, improves turbine reliability, and allows engineers to fine-tune upstream processes, boosting overall production yield.

Source: [Google Cloud's Customer Story](#)

Use Case: BMW – AI-Driven Visual Quality Inspection

BMW GROUP



How AI Helps

High-resolution cameras capture every vehicle as it leaves each station. **Convolutional neural networks**, trained on thousands of defect images, automatically flag paint imperfections, missing fasteners, or misaligned parts in real time. The system learns continuously from new data so detection accuracy keeps improving.

BMW uses AI to improve engineering processes, streamline purchasing, and enhance customer experience through chatbots, in addition to identifying defects.



The Problem

BMW's vehicle plants must detect tiny paint flaws and assembly defects on fast-moving production lines. Manual inspection is slow, labour-intensive, and inconsistent, leading to costly rework or warranty claims if defects slip through.

Source: [BMW Group](#)



Business Impact

BMW reports faster inspections, a significant reduction in rework, and more consistent quality standards across plants. Early detection prevents downstream defects, lowering warranty costs and improving customer satisfaction.

Use Case: Foxconn – Digital Twins & AI-Powered Quality Inspection



How AI Helps

Foxconn leverages AI in two main ways to optimize manufacturing. It uses **digital twins**, or virtual factory models, to simulate and perfect processes like machine scheduling and energy use before making physical changes. At the same time, it deploys **AI-powered computer vision** on production lines to quickly and consistently check for defects, like misaligned labels or improper grease application, a task that is far faster and more reliable than manual inspection.



The Problem

Foxconn operates in a low-margin electronics manufacturing sector facing tight cost pressures, rising energy bills, supply-chain challenges, and the need for high product quality. As demand grows—especially in AI servers and EV components—Foxconn needed to scale operations efficiently while keeping defects low, energy usage sustainable, and production workflows optimised.

Source: [emerj](#)



Business Impact

The adoption of digital twin modelling and AI inspection yields several benefits: reduced waste and scrap from defects, lowered rework cost, and improved yield. Energy consumption is optimised (reports suggest >30% reduction in energy usage in certain divisions), helping margins especially for energy-intensive product lines (like AI server production). Quality defects drop, leading to fewer returns or customer complaints, and throughput improves as bottlenecks are anticipated and addressed before they cause delays. Overall, these AI applications help Foxconn strengthen profitability, scale more sustainably, and improve consistency in product quality.

Use Case: Smart Maintenance Scheduling Using Existing Data

The Problem

Most manufacturing companies already track machine running hours, production output, and maintenance records, but they use this data only for basic reporting. Traditional maintenance either follows fixed schedules (whether machines need it or not) or waits for breakdowns to happen. Fixed schedules waste money on unnecessary maintenance, while reactive maintenance causes expensive downtime and emergency repairs.

How AI Helps

AI software analyzes the maintenance data you already collect from your existing systems without requiring new equipment. The technology studies patterns in machine performance, maintenance history, and production schedules to **predict when each piece of equipment will likely need service**. Instead of servicing machines every 30 days regardless of usage, AI creates personalized maintenance schedules for each machine based on its actual condition and usage patterns.

Generative AI creates easy-to-read maintenance reports and work orders that explain exactly why each machine needs service and what will happen if maintenance is delayed. The system can generate detailed instructions for maintenance tasks and create supply lists for parts that will be needed.

Business Impact

AI for predictive maintenance uses data and machine learning to **optimize maintenance schedules, reduce costs, and prevent equipment failures**. Rather than reacting to breakdowns or adhering to rigid schedules, predictive maintenance optimizes maintenance activities, ensuring that machines are serviced only when necessary. Planned downtimes decreases the risk of business not meeting manufacturing target, that may lead to huge liability due to contractual obligations.

Use Case: Intelligent Production Planning and Scheduling

The Problem

Creating production schedules that balance customer demands, machine availability, and material supply is extremely complex. Traditional scheduling often leads to bottlenecks where some machines sit idle while others are overloaded. This results in missed delivery dates, overtime costs, and inefficient use of equipment and workers.

How AI Helps

AI scheduling software uses your existing production data, order information, and machine capabilities to create optimized production schedules automatically. The system considers multiple factors simultaneously, such as machine setup times, worker skills, material availability, and delivery deadlines to find the best possible schedule.

When rush orders come in or machines break down, the AI can **instantly recalculate the entire schedule** to minimize disruptions.

Generative AI creates clear explanations for why certain jobs are scheduled at specific times and generates reports that help managers understand the reasoning behind scheduling decisions.

Business Impact

Companies using AI production scheduling typically **see improvements in on-time delivery rates** and **reductions in overtime costs**. The system helps maximize machine utilization while minimizing setup times and changeovers. This leads to higher productivity without requiring additional equipment or staff.

Use Case: Quality Control Data Analysis and Root Cause Detection

The Problem

Manufacturing companies collect lots of quality control data through testing and inspections, but analyzing this information to find patterns and root causes of defects is time-consuming and often missed entirely. Quality problems are usually addressed after they occur rather than prevented, leading to waste, rework, and customer complaints.

How AI Helps

AI quality analysis software processes your existing quality control data to identify patterns that **predict when defects are likely to occur**. The system can analyze test results, inspection reports, and production parameters to find connections between process conditions and quality problems that human analysis might miss.

With GenAI, manufacturing companies can conduct **continuous defect monitoring** and **root cause analysis (RCA)** for real-time quality control to ensure a more efficient and reliable production process. Generative AI creates detailed reports explaining what causes specific quality issues and recommends process adjustments to prevent problems before they happen.

Business Impact

AI quality control in manufacturing is the use of artificial intelligence to process information and automate tasks involved in taking care of quality during production. Companies typically see **reductions in defect rates** and significant **decreases in waste and rework costs**, due to catching deviations early using Artificial Intelligence. Such early detection of quality trends helps prevent major quality issues that could affect entire production batches.

Use Case: Automated Documentation and Compliance Reporting

The Problem

Manufacturing companies must maintain extensive documentation for quality standards, safety regulations, and customer requirements. Creating reports, updating procedures, and maintaining compliance documentation takes significant time away from production activities. Many companies struggle to keep their documentation current and comprehensive.

How AI Helps

Generative AI can **automatically create and update manufacturing documentation** by analyzing your production data, quality records, and compliance requirements. The system can generate standard operating procedures, quality reports, and compliance documentation that meets industry standards and regulatory requirements.

Generative AI creates new content like text, images and code by learning patterns from data and previous prompts. In industry, it has a variety of uses for product searches, document summarization, customer service, call processing and more. The technology can also **create training materials** for workers and generate clear explanations of complex procedures in multiple languages if needed.

Business Impact

Companies using AI documentation systems typically **reduce the time spent on paperwork** while improving the quality and consistency of their documentation. This allows technical staff to focus more time on production and problem-solving activities. Better documentation also improves training efficiency and helps maintain consistent quality standards across different shifts and workers.



Thank you



All Around

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